

Energy and Power Load Forecasting Using Machine Learning

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Abstract—Predicting energy and power loads is crucial for effective power system planning, operation, and management. This project aims to use data science and machine learning to forecast future energy demand using past consumption data. Proper load forecasting can optimize energy distribution, lower operational costs, and enhance system reliability.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Accurate energy load forecasting is now crucial for modern power systems due to the rising demand for electricity. The goal of this research is to accurately predict future power use using past energy data and machine learning techniques.

II. LITERATURE REVIEW

Previous research have used statistical and machine learning methods, including linear regression, time series models, and neural networks, to forecast energy loads. These methods highlight how data-driven techniques can improve forecast accuracy.

III. METHODOLOGY

This project uses historical consumption of electricity data to train and evaluate machine learning models. To increase prediction accuracy, data preprocessing procedures like normalization and feature extraction are used prior to training the model.

IV. RESULTS

Forecasting models are evaluated based on error metrics like mean absolute error and root mean square error. The study should demonstrate if machine learning algorithms can accurately capture consumption trends in energy load data.

V. DISCUSSION

The results indicate that machine learning approaches are suitable for energy and power load forecasting. Variations in prediction accuracy may be influenced by factors such as data quality, seasonal patterns, and external variables like weather conditions. Improving feature selection and incorporating additional contextual data could further enhance forecasting performance.

VI. CONCLUSION

This research will show how data science and machine learning can effectively estimate energy and power loads. Future studies may incorporate employing better deep learning models and adding weather data.

REFERENCES

- [1] Y. Okafor, "Energy and power load forecasting using machine learning (source code)," GitHub repository, 2025. [Online]. Available: https://github.com/yvokafor-rgb/DS_ML_Project_colab-integration
- [2] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.
- [3] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [4] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [5] K. Elissa, "Title of paper if known," unpublished.
- [6] R. Nicole, "Title of paper with only first word capitalized," *J. Name Stand. Abbrev.*, in press.
- [7] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [8] M. Young, *The Technical Writer's Handbook*. Mill Valley, CA: University Science, 1989.